

Ahsan Farabi

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LinkedIn — GitHub — Google Scholar — Portfolio

RESEARCH INTERESTS

Vision-Language Models (VLMs), Large Language Models (LLMs), Human-Computer Interaction (HCI), AI Safety, AI for Environmental Sustainability

EDUCATION

United International University , Dhaka, Bangladesh B.Sc. in Computer Science & Engineering, Major: Data Science	2022 – 2026 CGPA: 3.88/4.00
Uttara Model College , Dhaka, Bangladesh Higher Secondary Certificate (HSC), Group: Science	2018 – 2020 GPA: 5.00/5.00
Uttara Model School , Dhaka, Bangladesh Secondary School Certificate (SSC), Group: Science	2016 – 2018 GPA: 5.00/5.00

TEACHING EXPERIENCE

Undergraduate Teaching Assistant <i>United International University</i> Courses: Object-Oriented Programming Lab, Database Management Systems Lab	Onsite June 2024 – October 2025
- Assisted in delivering laboratory sessions, technical demonstrations, and weekly consultation hours for 100 undergraduate students across two laboratory courses. - Designed supplementary learning materials, programming exercises, and exam preparation resources to reinforce core concepts and improve student learning. - Mentored student project teams throughout the semester; supervised a team that won the <i>Champion</i> award at the UIU CSE Project Show.	

RESEARCH EXPERIENCE

WADE: A Waste Annotation Reasoning-Chain Dataset and Distillation Framework for Low-Resource Vision-Language Models

Submitted to Engineering Applications of Artificial Intelligence (EAAI), Elsevier — Undergraduate Thesis [code]
Supervised By: Dr. Muhammad Nomani Kabir

- Developed a multimodal dataset comprising 2,167 in-situ images of Bangladeshi water bodies with 13,608 manually annotated waste instances and reasoning chains using a custom annotation platform.
- Conducted a comprehensive benchmark of six vision-language models under zero-shot, few-shot, and reasoning-based prompting, identifying perception—not reasoning—as the primary limitation in aquatic waste understanding.
- Fine-tuned an open-source vision-language model using QLoRA, improving domain-specific perception while reducing hallucinations in complex aquatic environments.

GreenSwap: Verb Choice or Output Length? A Controlled Decomposition of Prompt-Level Energy Savings in LLM Inference

Under review, Empirical Methods in Natural Language Processing (EMNLP) [code]
Supervised By: Shahriar Mahmud

- Tested whether changing prompt wording or limiting response length saves more energy during LLM inference, across 21,600 test runs.
- Found that telling the model to keep answers short saves more energy and gives better quality than swapping words in the prompt.

EcoDetect: Lightweight Real-Time Recyclable Waste Detection Using YOLOv11n Trained on LLM-Guided Synthetic Data

Published, International Conference on Computer and Information Technology (ICCIT), IEEE Xplore [code] [paper]
Supervised By: Md. Tanvir Raihan

- Used AI-generated synthetic images to train a lightweight waste detection model, avoiding the need for manual data collection.
- Achieved 78.9% precision and 81.1% mAP@0.5 on real-world test images the model had never seen before.

SELECTED PROJECTS

Orchestra (Local AI Coding Agent) [github] [download]	2026
<i>Technology: Python, Ollama, TUI</i>	
Built a privacy-first, terminal-native AI coding agent running entirely on local Ollama models, with autonomous tool-calling for file I/O, bash execution, and web search, isolated multi-session context, and a permission-gated safety layer for destructive operations.	
Wiselearn (Python Library) [github] [download]	2026
<i>Technology: Python, PyTorch, HuggingFace</i>	
Built an educational machine learning library that streamlines dataset preparation, model training, and explainability workflows for students learning ML fundamentals.	
Dispatch (API Testing Tool) [github] [live]	2026
<i>Technology: JavaScript, Next.js, Tailwind CSS, Zustand</i>	
Built a browser-based API testing platform supporting workflow automation, authentication handling, and endpoint validation.	
EcoRover (Waste Sorting Robot) [github] [video]	2025
<i>Technology: Python, OpenCV, YOLOv8n, Raspberry Pi 5, Arduino, Servo Motor Control</i>	
Built a 4-DOF robotic system that autonomously classifies and sorts recyclable waste in real time using computer vision.	
UIU Platformer (Onboarding Game) [github] [video]	2024
<i>Technology: Java, JavaFX</i>	
Built a 2D onboarding simulation game for incoming students and evaluated its usability through testing with 20+ participants.	

AWARDS & HONORS

Finalist - National Blockchain Olympiad Bangladesh	2025
On-Campus Champion - Hult Prize UIU (Social Entrepreneurship Competition)	2026
1st Runner-Up - UIU CSE Project Show (DBMS Poster Award)	2023
4th Runner-Up - UIU CSE Project Show (Database Management Systems)	2023
2nd Runner-Up - UIU CSE Project Show (System Analysis & Design)	2024
2nd Runner-Up - UIU CSE Project Show (Advanced Object-Oriented Programming)	2024
4th Runner-Up - UIU CSE Project Show (Software Engineering)	2025
Merit Scholarship - United International University	2022 – 2026

CERTIFICATIONS

Machine Learning Specialization - Stanford University [certificate]	2024
Git & Github - Google [certificate]	2024
Raspberry Pi Essential Training - NASBA [certificate]	2025
Scientific Computing with Python - freeCodeCamp [certificate]	2026
Claude Code 101 - Anthropic [certificate]	2026

SKILLS

Programming Languages: C, C++, Java, Python
Scripting Languages: BASH, HTML, PHP, CSS, JavaScript, SQL, LaTeX
Version Control: Git, GitHub
Design Tools: Proteus Circuit Simulator, Logisim Circuit Simulator, CISCO Packet Tracer
AI & Machine Learning: NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, Roboflow, RAG, LangChain, ADK
Web Development: Next.js, Tailwind CSS, Supabase, Vercel
Hardware: Raspberry Pi, ESP32, Arduino
Tools: MS Word, PowerPoint, Excel, Jira, Canva
Soft Skills: Communication, Leadership, Teaching, Design Thinking, Public Speaking, Curriculum Design

REFERENCES

Dr. Muhammad Nomani Kabir

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Dr. Mohammad Nurul Huda

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